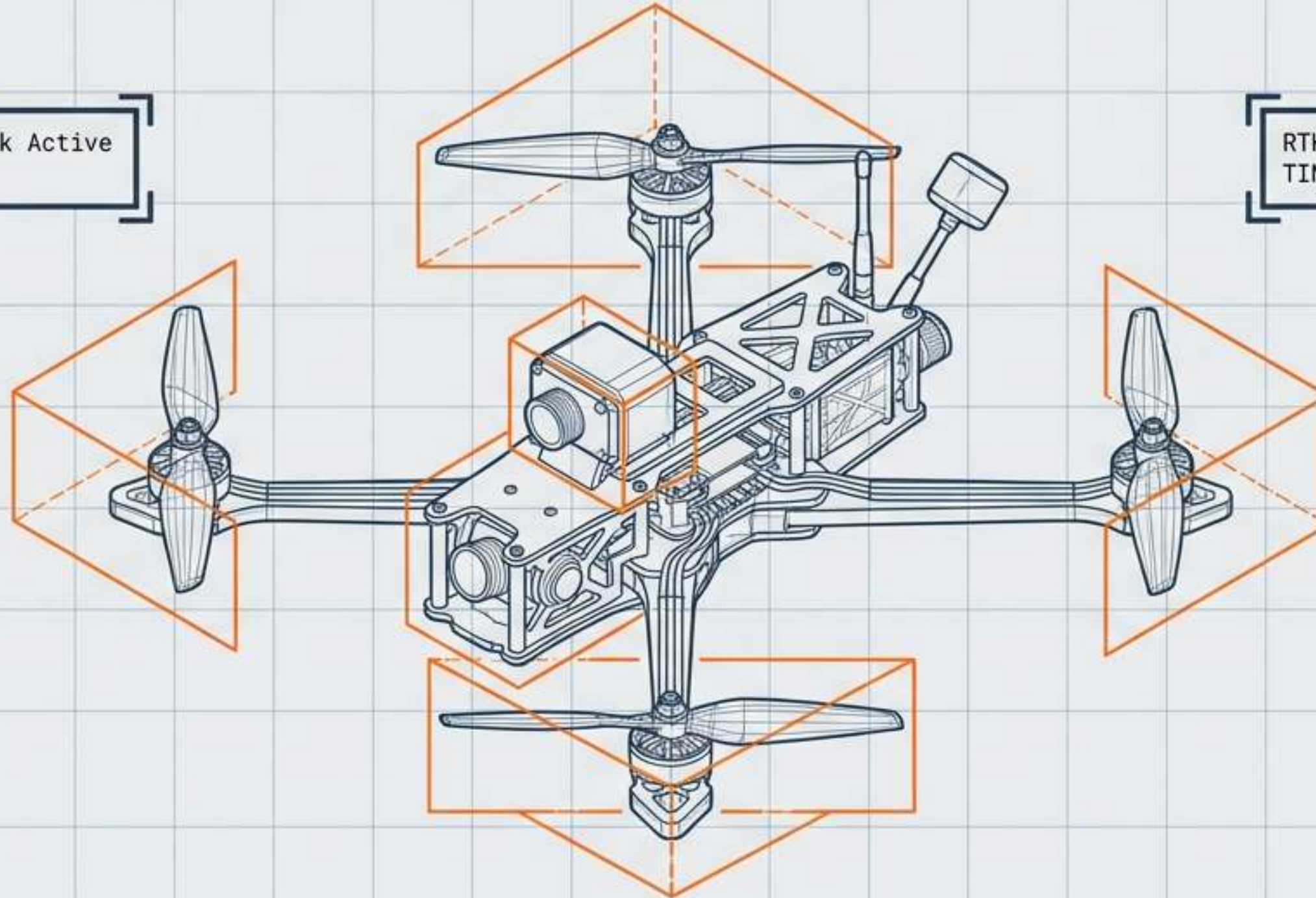


COMPUTATIONAL SWARM ENGINEERING

Advanced UAV Systems & The Architecture of Aerial Entertainment

SYSTEM INIT: UATG Framework Active
LATENCY: < 1.5s

RTK-GNSS: Centimeter-Level Lock
TIMING: Sub-microsecond PPS



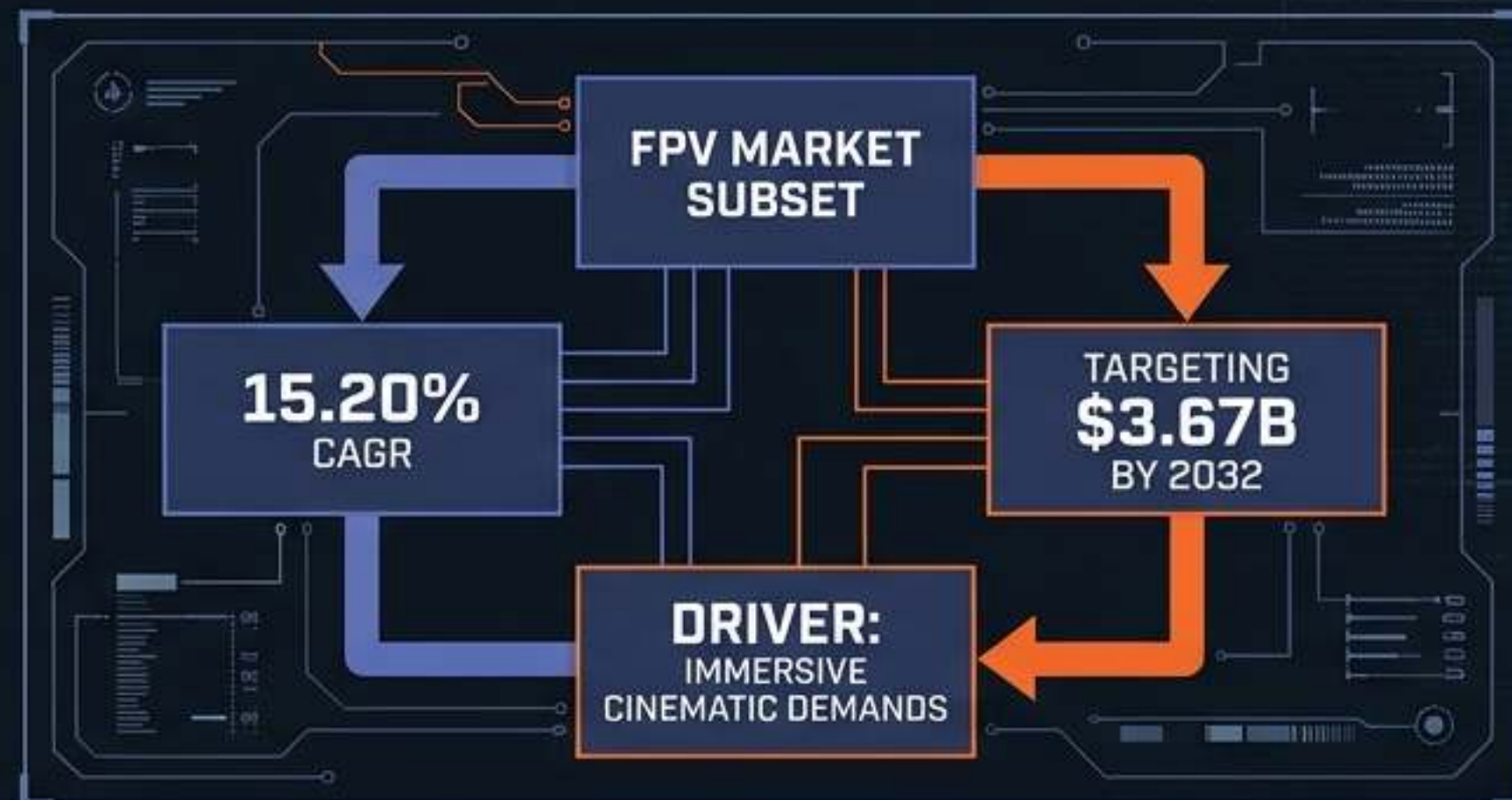
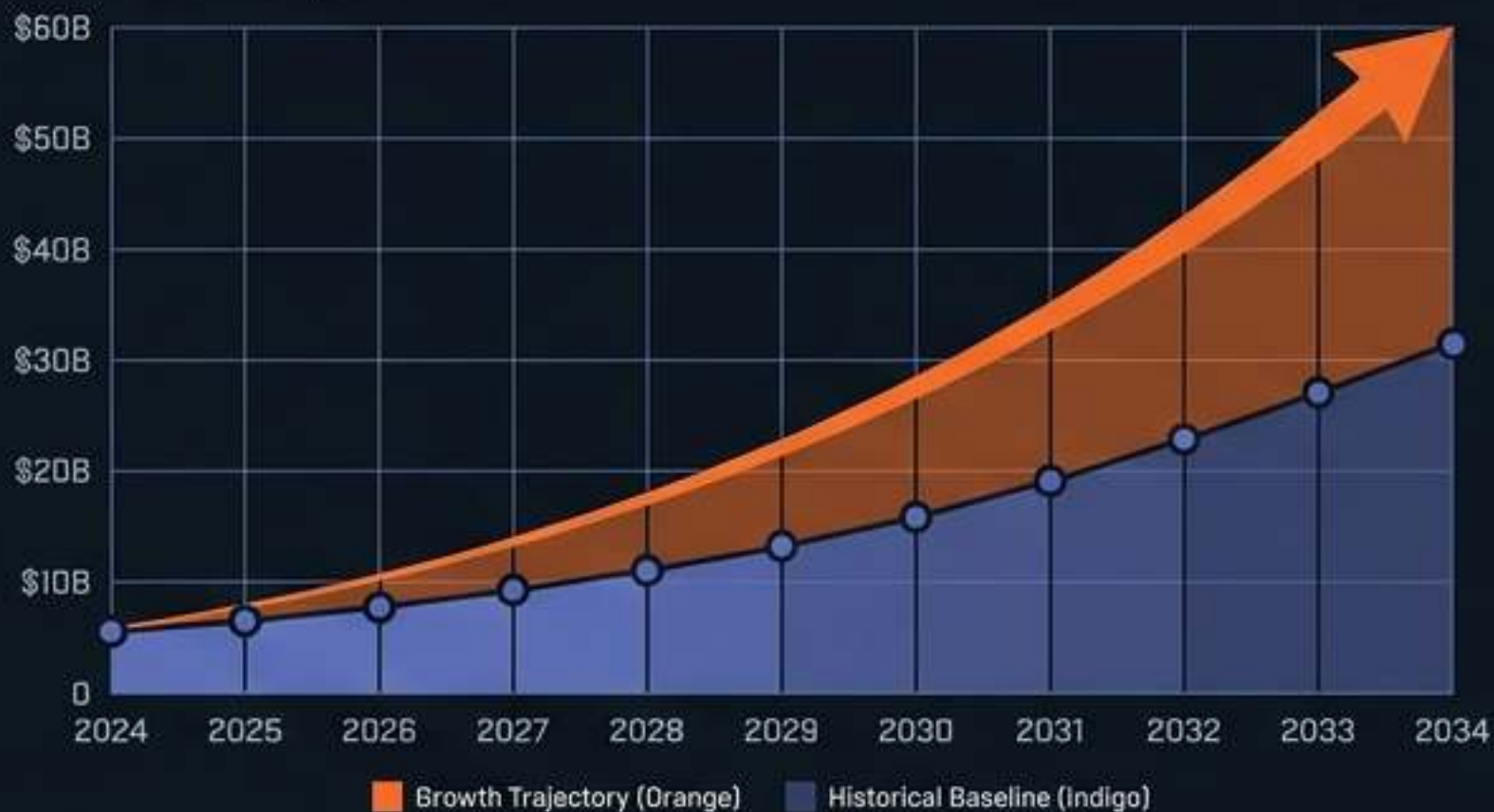
SECTOR TAXONOMY 4.6: GLOBAL MARKET TRAJECTORIES

16.5% SWARM
CAGR

PROJECTED
\$58.6B 2034
VALUATION

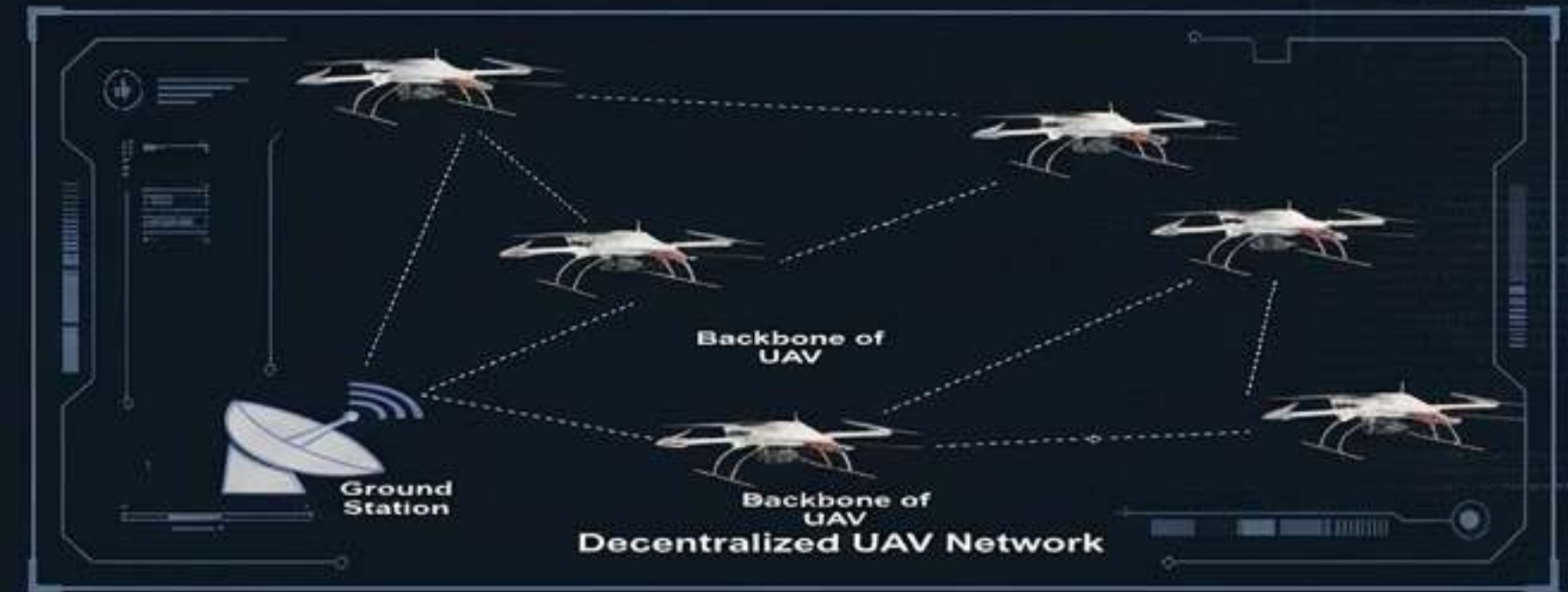
SWARM SYSTEMS MARKET GROWTH TRACKER

Real-time Telemetry Data



ARCHITECTURAL PARADIGMS OF SWARM INTELLIGENCE

DATE: 2024-10-27 (TIME) 14:00
FILE: SWARM_ARCHITECTURE_DIAGRAMS_V1.0
AUTH: DR. ALEXANDER V. KOSTIN
DESC: ARCHITECTURAL PARADIGMS OF SWARM INTELLIGENCE
TAGS: SWARM, UAV, INTELLIGENCE, ARCHITECTURE
NOTE: KOSTIN, ALEXANDER V. 2024-10-27



CENTRALIZED

- **Mechanism:** Single Ground Control Station dictates all paths.
- **Advantage:** Simplified path planning for structured light shows.
- **Constraint:** Single point of failure.

DECENTRALIZED

- **Mechanism:** Relies on local interactions and biological swarm logic.
- **Advantage:** Fault tolerance and scalability.
- **Constraint:** Requires sophisticated local onboard logic.

DISTRIBUTED

- **Mechanism:** No central leadership; peer-to-peer decision making.
- **Advantage:** Maximum adaptability in dynamic environments.
- **Constraint:** High communication and bandwidth overhead.

LEADER-FOLLOWER (HYBRID)

- **Mechanism:** Designated master UAVs relay GCS commands to followers.
- **Advantage:** Balances network stability with infrastructure-less capabilities.
- **Constraint:** Network partitioning risk if leader fails.

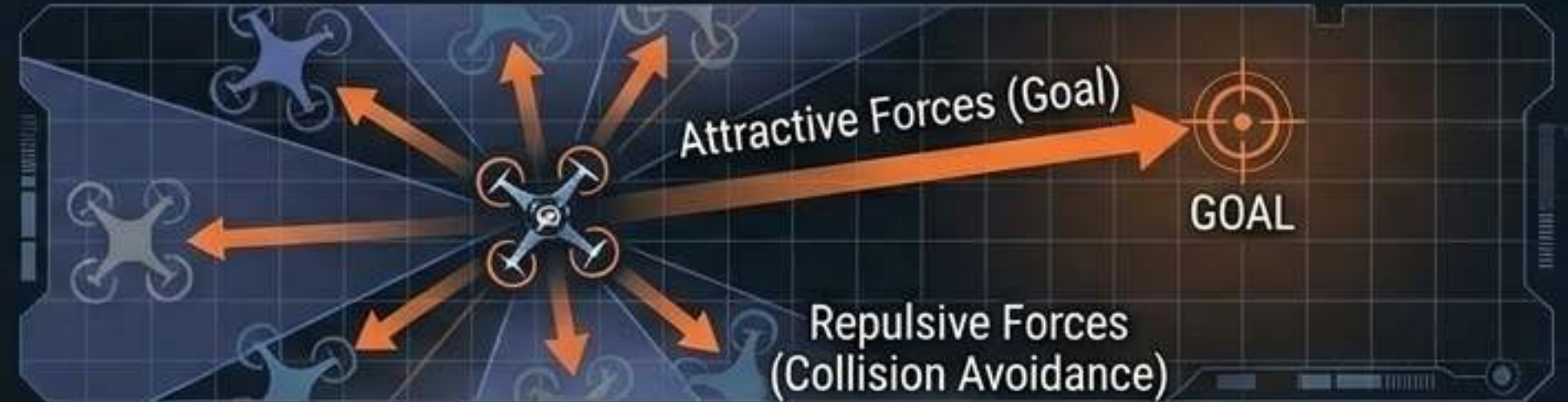
Swarm Coordination Algorithms & Emergent Logic

Consensus-Based Algorithms

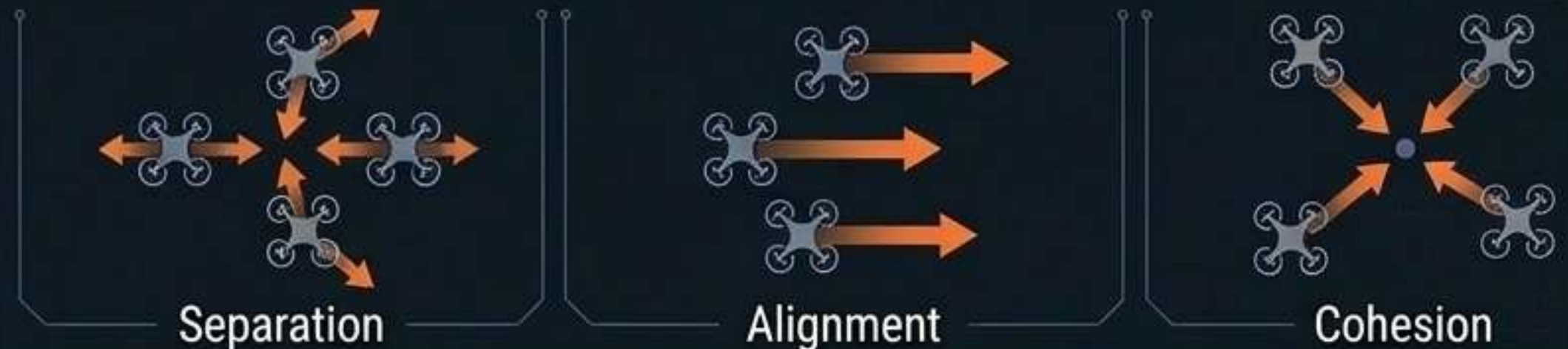
Focuses on state agreement (position, velocity) over time. Graph theory defines influence weights between neighboring drones.

$$x_i(t+1) = x_i(t) + \sum_{j \in \mathcal{N}_i} w_{ij}(x_j(t) - x_i(t))$$

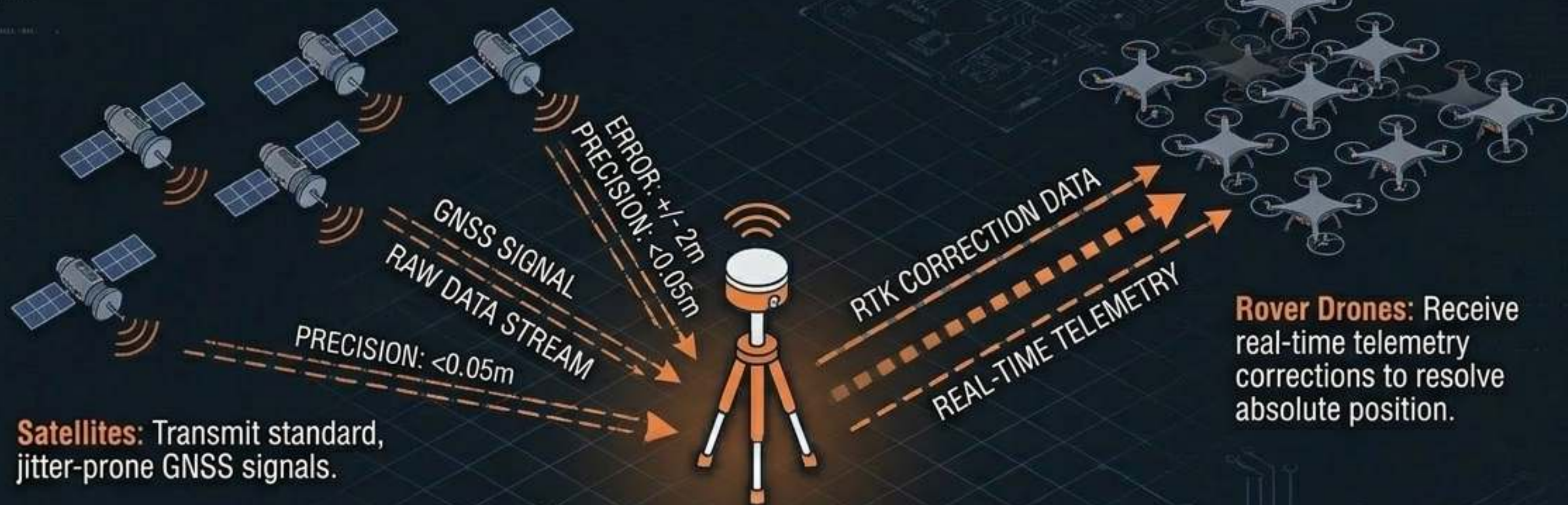
Potential Field Methods



Rule-Based Logic (Boids)



HIGH-PRECISION POSITIONING: RTK-GNSS MECHANICS



Satellites: Transmit standard, jitter-prone GNSS signals.

Static Base Station: Known surveyed position calculates carrier-phase errors and atmospheric interference to generate correction dataset.

Rover Drones: Receive real-time telemetry corrections to resolve absolute position.

REQUIREMENT: CENTIMETER-LEVEL PRECISION

System enables non-negotiable 0.3 to 0.4 meters flight separation without visual jitter.

TEMPORAL SYNCHRONIZATION & CLOCK DISCIPLINING

For a swarm to act as a single mathematical entity, flight controllers and LED firing sequences must share a unified master clock, independent of network delays.

NTP (Network Time Protocol)

- ✈ Standard computer sync.
- 🕒 **Accuracy:** Millisecond.
- 🔧 **Drawback:** Vulnerable to non-deterministic wireless latency and packet jitter over ad-hoc networks.

PTP (Precision Time Protocol / IEEE 1588)

- ✈ LAN-based sync.
- 🕒 **Accuracy:** Microsecond.
- 🔧 **Application:** Hardware timestamping in structured networks, used in local control clusters.

GPS-PPS (Pulse-Per-Second)

- ✈ The Aerospace Standard.
- 🕒 **Accuracy:** Sub-microsecond (10 billionths of a second).
- 🕒 **Mechanism:** Disciplines internal clocks using atomic satellite time via a Mobile Edge Clock Server (MECS), reducing sync errors by 99.99%.

Synchronized.
high precision

COMPUTATIONAL GEOMETRY: THE UATG FRAMEWORK

TARGET ALLOCATION

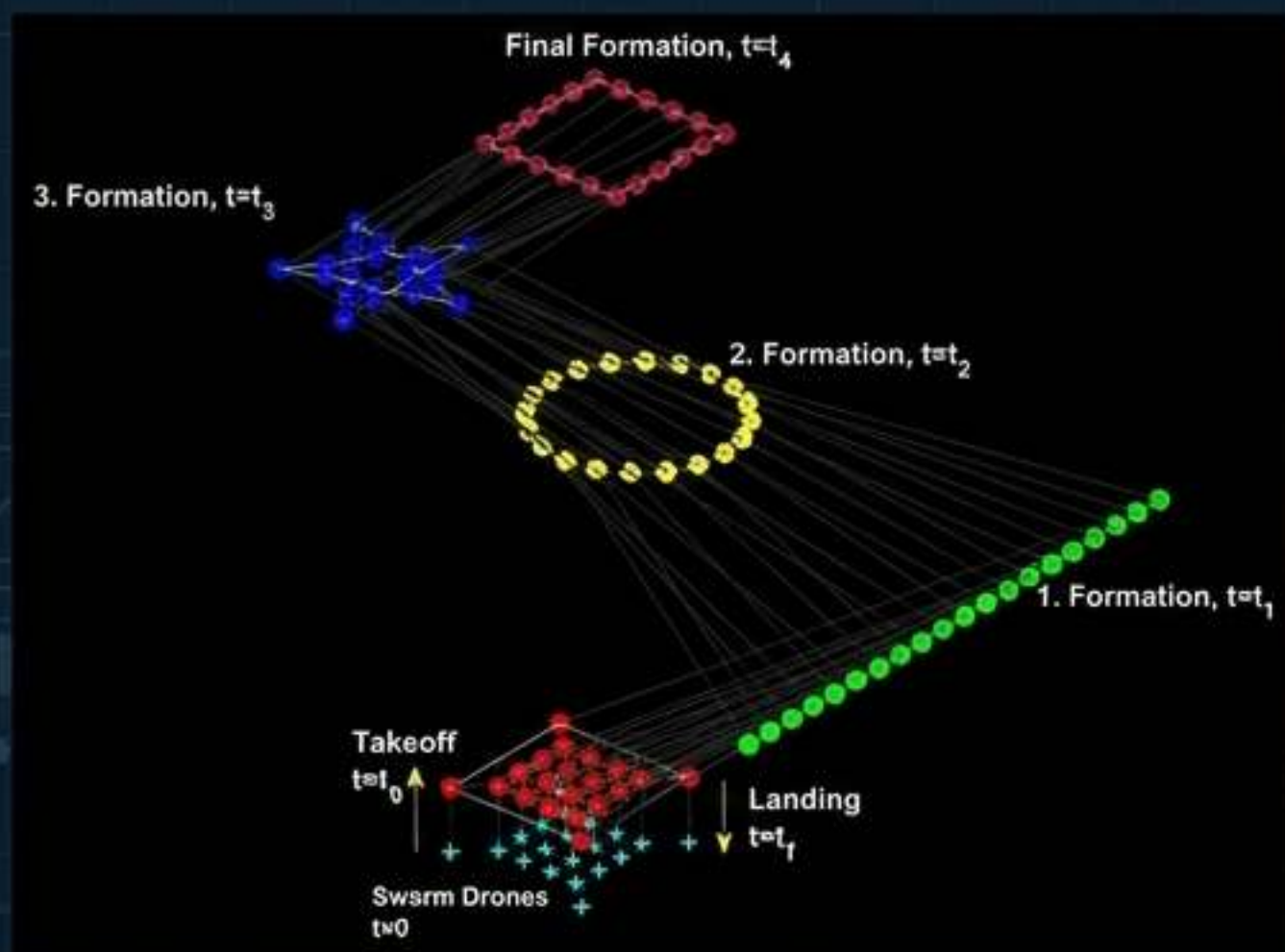
Solves the Linear Assignment Problem using the Hungarian Algorithm to map N drones to M waypoints.

TRAJECTORY GENERATION

Computes optimal, dynamically feasible piecewise polynomials while enforcing actuator limits.

COLLISION VERIFICATION

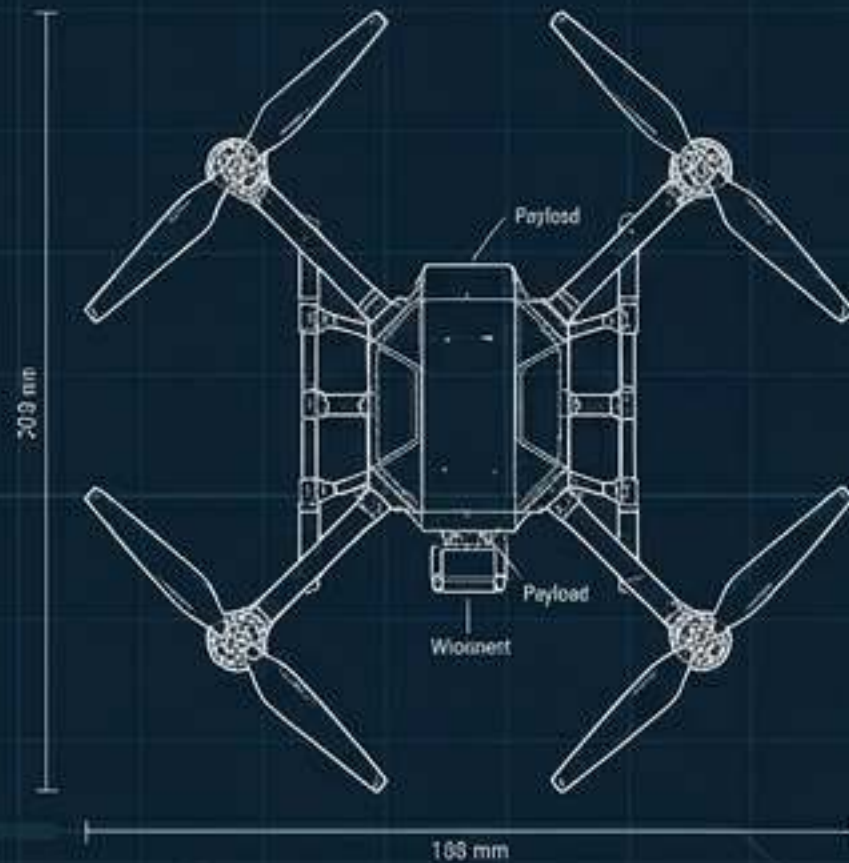
Re-planning loops perturb waypoint timings to resolve detected drone-to-drone conflicts.



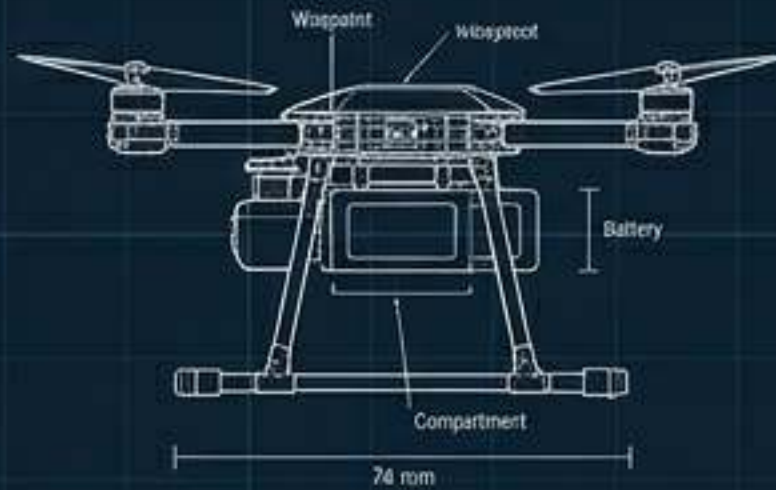
PERFORMANCE BENCHMARK:
Computes optimal trajectories for 1,008 drones in ~ 1.5 seconds on a standard laptop.

AIRFRAME & PAYLOAD ENGINEERING SPECIFICATIONS

CROSTARS C5 (HEAVY-DUTY)

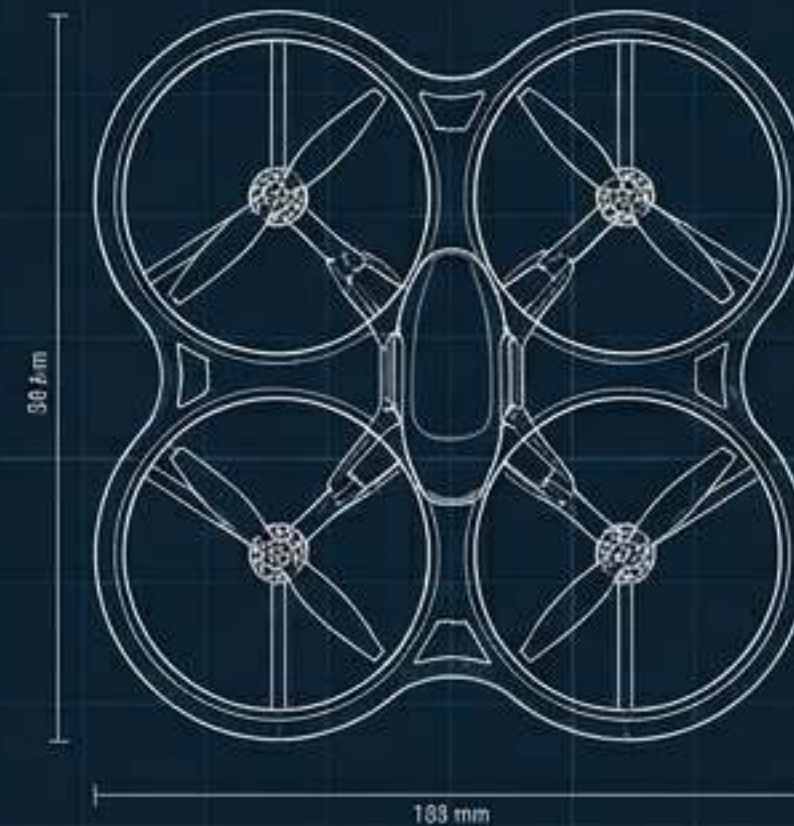


Weight: 630g
Wind Resistance: 13.8 m/s (Level 6)
Lighting Power: 24W RGB



Speed: 8 m/s
Battery: 14.8V 4800mAh (Smart swap)
Waterproof: IP54

MAZZY STAR (LIGHTWEIGHT)



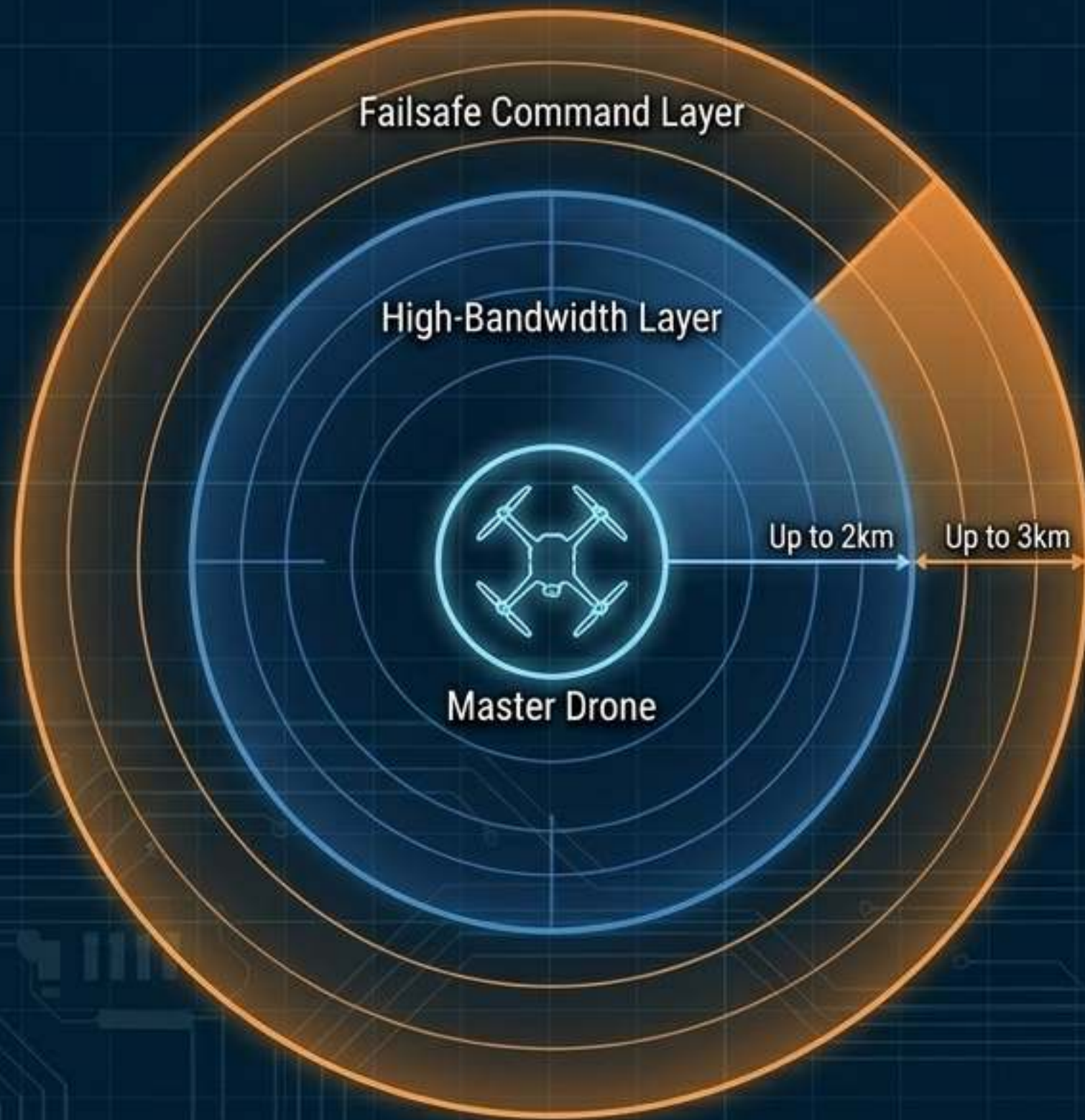
Weight: 480g
Wind Resistance: < 7 m/s
Lighting Power: 4W RGB



Speed: 5 m/s
Battery: 7.6V 3000mAh
Safety: Full-size propeller guards

OPERATIONAL REALITY: Hardware is heavily optimized for rapid turnaround (20-minute batch power cycles) and high-luminance LED visibility against urban light pollution.

MULTI-BAND COMMUNICATION TOPOLOGIES



HIGH-BANDWIDTH LAYER: WiFi 6

Frequencies: 2.4 GHz, 5.2 GHz, 5.8 GHz

Range: Up to 2km

Function: Heavy data loads, high-bandwidth telemetry, show files, and visual choreography.

FAILSAFE COMMAND LAYER: Sub-GHz LoRa

Frequencies: 433 MHz / 850-930 MHz

Range: Up to 3km

Function: Extremely robust, penetrates smartphone interference. Exclusive to essential command and emergency signals.

MLINK PROTOCOL ENGINE

Event-Driven Network

Publish-Subscribe Mechanism

Reduces Redundant Traffic & Throttling

DEPLOYMENT ENGINEERING: THE SURVEY-IN PHASE

1. RTK Base Survey-In

The tripod-mounted base station locks its absolute position. It must remain undisturbed to provide millimeter-level corrections.

2. Grid Initialization

Drones are placed in Box Mode. The 0.3m spacing achieves 80% space efficiency—fitting 1,000 drones in just 200 sq meters.



3. RTK Fixed State

Ground Control Station uploads the show project. Drones independently verify centimeter-level lock.

4. Set Home & Launch

The global home point is synchronized across the fleet, allowing the swarm to ascend simultaneously into formation.

THE 5 PILLARS OF SWARM SAFETY ARCHITECTURE

1. Navigation Redundancy

Up to six independent estimators prevent sensor failure and data inaccuracy, stopping flyaways.

2. Up-to-Date Software

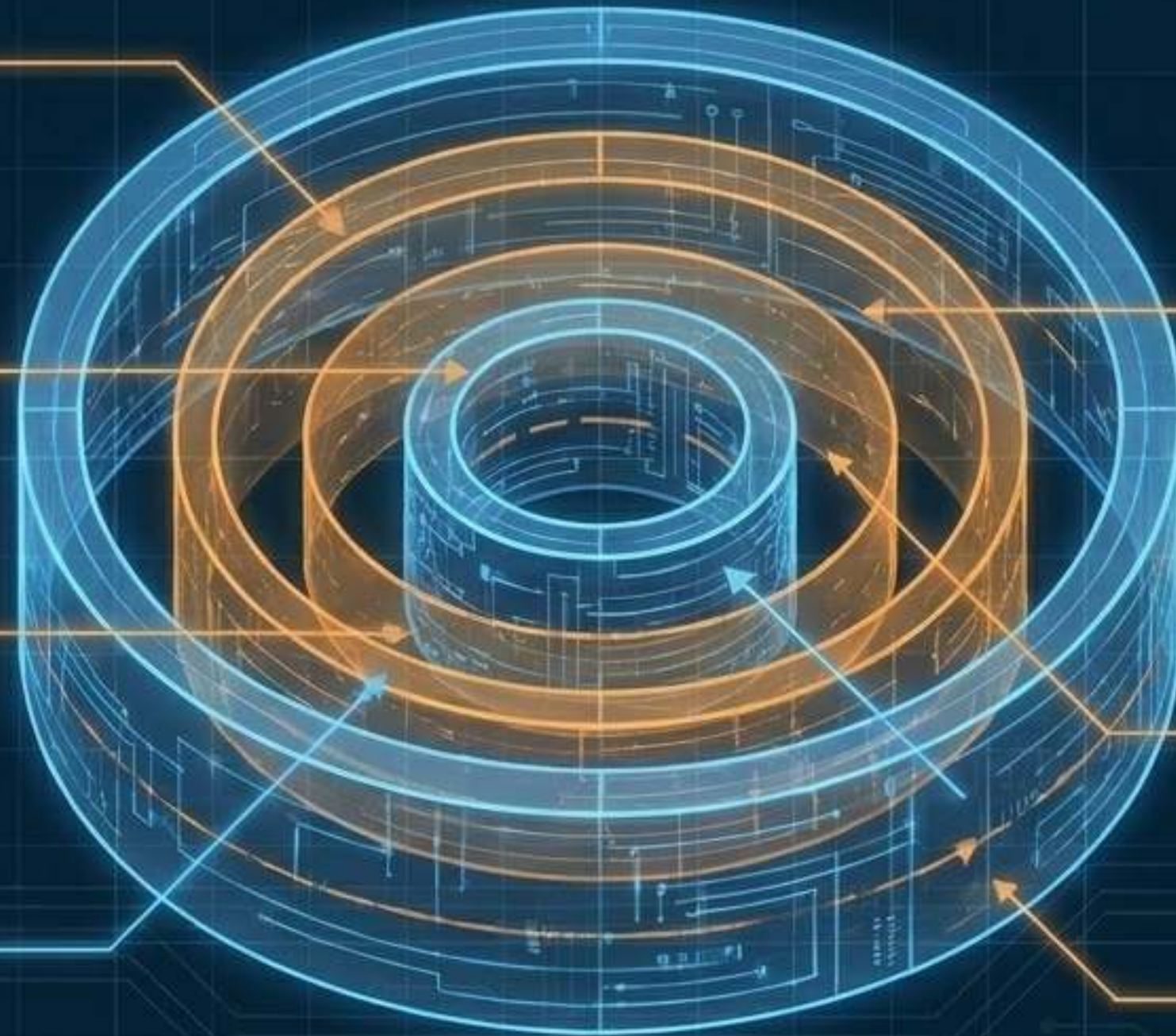
Utilizing PX4 v1.15+ for onboard mathematical flight deconfliction.

3. Multi-Layer Geofencing

Three boundaries: soft dynamic bubbles, hard geofences, and predictive trajectory fences.

1. Navigation Comms

Up to six ensemble estimators entimety, stopping flyaways.



AUTOMATED FAILSAFE TRIGGERS

- **GPS Loss:** Slow automated descent.
- **Low Battery:** Return to Launch (RTL) sequence.
- **Geofence Breach:** Immediate termination.

4. Long-Range Comms

bypassing congested WiFi for emergency control.

5. Safety Management (SMS)

Commercial aviation-grade pilot training and procedural discipline.

REGULATORY EVOLUTION & OPERATIONAL NORMALIZATION

FAA Part 108 (Federal)

Impending rule to normalize Beyond Visual Line of Sight (BVLOS) operations and legally codify the operation of multiple aircraft by a single pilot.

MLV (Operator)

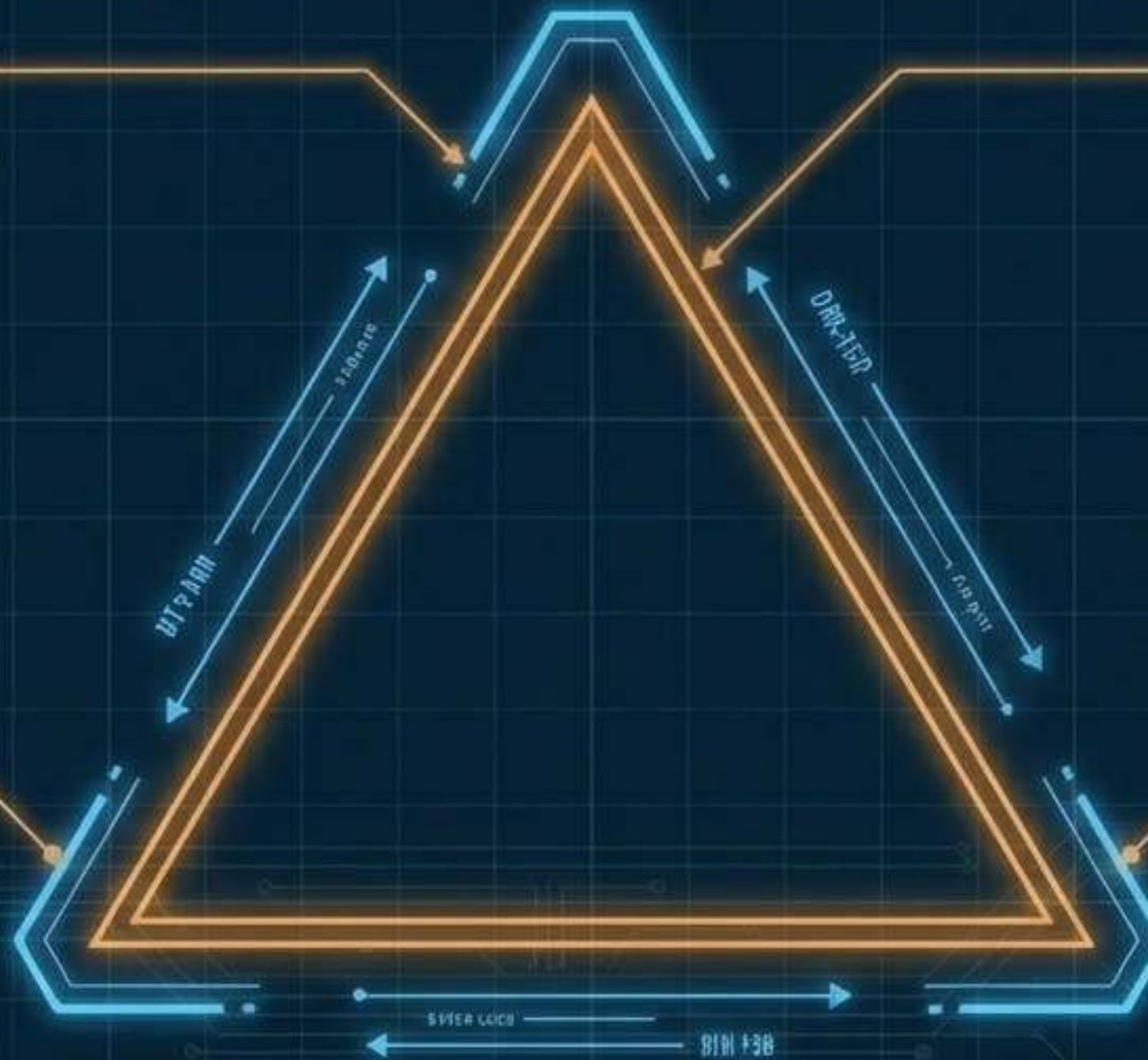
Drones consortium committee on monetization, communication and cybersecurity consultation.

ASTM F38 (Manufacturer)

Global standards committee developing rigorous airworthiness and hardware production standards for swarm-capable UAVs.

DLSA (Operator)

Drone Light Show Alliance. Industry consortium providing immediate, platform-agnostic safety baselines and peer-reviewed incident analysis before federal rules finalize.

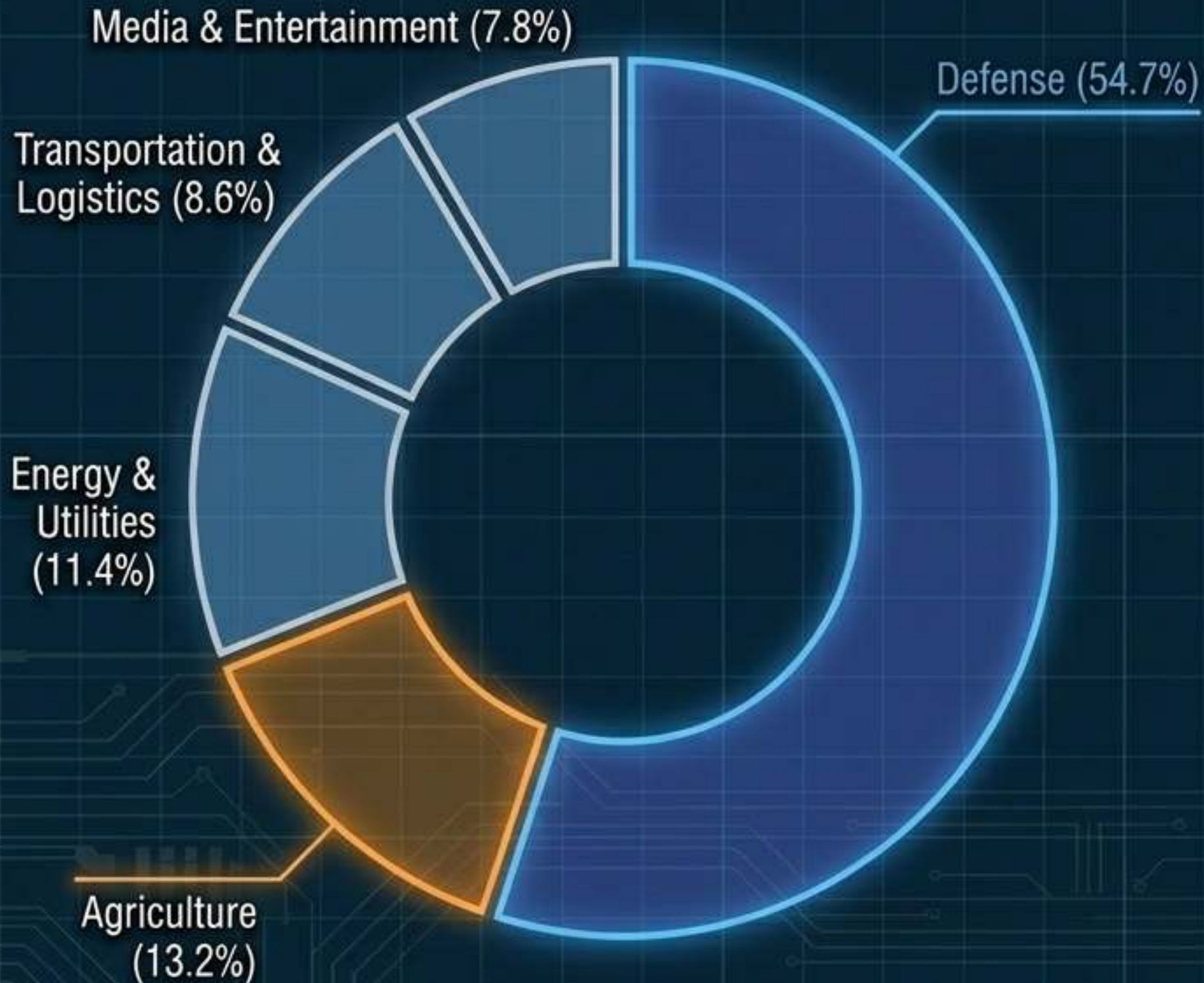


Decentralized Fault-Tolerant Architectures

Scenario: Distributed Data Aggregation (DDA) - A leader drone or network partition fails.



CROSS-SECTOR SWARM PROPAGATION



Defense (54.7%)

The dominant end-user. Utilizing decentralized swarms for autonomous tactical reconnaissance and counter-UAS tracking.

Agriculture (13.2%)

Fastest Growing (22.1% CAGR). Swarm sprayers cover 50-100 acres/hour (vs 15-20 for single units).

Energy & Utilities (11.4%)

Coordinated inspection swarms utilizing LiDAR and gas-spectrometry to predict pipeline corrosion.

Transportation & Logistics (8.6%)

Urban air mobility corridors and heavy-lift cargo pods.

Media & Entertainment (7.8%)

The highly visible incubator stress-testing the tech stack for all other sectors.

Conclusion: The Engineering Legacy of Aerial Spectacle

Visual Demand

Perfectly smooth, non-jittering 3D shapes.



Thousands of drones moving simultaneously.



Performing safely over dense crowds.



Engineering Reality

Sub-microsecond GPS-PPS clock disciplining and RTK-GNSS centimeter-level corrections.

UATG mathematical trajectory optimization and Multi-band WiFi6/LoRa networking.

Decentralized fault-tolerance, dynamic geofencing, and multi-estimator redundancy.

The entertainment drone sector is not merely a commercial novelty; it is the most aggressive, high-visibility R&D incubator for autonomous, fault-tolerant aerospace systems operating in the national airspace.